Elijah Ifechukwu Nelson

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Summary

Education

M.Phil. in Machine Learning and Machine Intelligence, University of Cambridge10/2025 - 08/2026

- Advanced Probabilistic ML, Optimization & Reinforcement Learning.
- MPhil thesis — Cambridge, supervised by Andrew Fitzgibbon (Former Microsoft Research), Investigating whether structured (Kronecker-factored, low-rank) approximations of the transformer Hessian can yield optimisers that match or exceed Muon, Shampoo, and SOAP on language-model pretraining. Includes from-scratch implementations of Muon and Shampoo, full-Hessian and Gauss–Newton spectral analysis on small transformers, and scaling-law evaluation following the Wen et al. (2025) fair-comparison protocol.

B.Eng. in Electrical and Electronics Engineering, Covenant University 08/2018 - 09/2023

- Grade: First Class Honors; CGPA: 4.98/5.0 (Highest in Department's History).
- Class rank: 1 of 132; Institute rank: 1 of 1,175 (Valedictorian).
- Thesis: Design, simulation, and implementation of a multi-purpose unmanned aerial vehicle (UAV) with autonomous navigation, precision landing, and visual tracking capabilities. [link]

Research Interests

ML + Systems, Embedded Systems, Embedded AI, Sensor Design, IoT, Embedded Software & OS, Computer Architecture, Robotics, Integrated Circuits.

Research & Teaching Experience

AI Research Assistant, Intelligent Manipulation Lab, University of Lincoln 03/2022 - 06/2023

- Conducted experiments on AI ensemble models and tactile data representations to advance slip classification in robotic manipulation. [link]
- Developed a novel SMOTE-CNN deep-learning technique for real-time slip classification on a limited, imbalanced dataset of temporal slip states without relying on kinematic state, achieving 98% slip detection accuracy and a 95% reduction in false negatives.

Research Assistant, Dept. of Electrical & Electronics Engineering, Covenant University 01/2023 - 07/2023

- Designed and developed several projects on embedded software for autonomous and IoT systems (e.g., automated waste management and autonomous navigation for a vacuum cleaning robot).
- Taught and trained over 50 students on research methodologies in embedded systems development.

Tutor, Covenant University 01/2021 - 07/2023

- Taught and provided personally curated notes and interactive case studies to more than 200 students in Electromagnetism, Applied Electronics, Advanced Control Systems (ACS), Power Electronics, and High Voltage Engineering.
- Achieved an average pass rate of 85% across all subjects, with a 77% pass rate in ACS.

Publications

[1] T. Somefun, N. Elijah, A. E. Igho-Orere, O. Timilehin, C. Somefun, and S. Ongbali. Energy optimization algorithm for reducing energy consumption in a smart home. In *2023 2nd International Conference on Multidisciplinary Engineering and Applied Science (ICMEAS)*, pages 1–5, 2023. [link].

[2] N. Elijah, K. Nazari, and A. Ghalamzam. Advancing slip classification in robotic manipulation through tactile data representation and model selection. In *International Conference on Control, Mechatronics and Automation (ICCMA)*, 2023. [link].

Professional Experience

Systems Engineer (Founding), Climate in Africa — 10/2023 - 08/2025

- Developed low-cost, WMO-compliant Modular In situ Reporting Instruments (MIRI) covering 14 meteorological parameters, in a mission to deploy 10,000+ units across Africa over 3 years, with presence in Nigeria, Ivory Coast, Kenya, and Ghana within the first 3 months.
- Developed a NOR Flash (MT25QL01GBBB) driver and co-designed CRC-verified OTA updates over BLE and octet-streams for resilient field updates; ported 3K LOC from ATmega to STM32 (F4/F103), yielding a 30% performance increase.
- Designed firmware for MIRI (3.5 days per charge) using a sleep scheduler, RTOS, and non-blocking state machines to reduce GSM/GPRS latency by ∼70%, with autonomous fallback logging to NOR Flash/SD card on connectivity loss.
- Developed a non-blocking task scheduler firmware for autonomous satellite tracking (azimuth and elevation actuation) and a self-designed LoRa-based base-to-ground-station for telemetry and control.
- Built the mail and notification architecture of climateinafrica.com aggregating TBs of data from MIRI monitors and users; architected a NOAA pipeline to scrape and parse GB-scale climate datasets for the analytics sandbox.
- Led the entire product documentation effort for climateinafrica.com.

Machine Learning Engineer, Ticki — 08/2024 - 07/2025

- Architected a scalable facial-similarity retrieval system using OpenCV, FAISS, and async Celery tasks, optimizing end-to-end latency by 70% (sub-100 ms) through in-memory caching and concurrent feature search across FAISS sub-indexes.
- Achieved near-zero downtime and high-throughput processing for 1,000+ concurrent user requests via a production-grade Ubuntu deployment with multiple Gunicorn workers, Nginx load balancing, and Redis-based deduplication.

Reservoir Performance Simulation Intern, Schlumberger — 04/2022 - 08/2022

- Executed maintenance measures and proposed innovative designs, resulting in a 15% efficiency boost and substantial cost reductions.
- Conducted 20+ test procedures to assess the integrity of various oil wells, contributing to enhanced operational efficiency.

Delegate of Croatia, Model United Nations, Beirut, Lebanon — 03/2019 - 04/2019

- Conducted research on Indigenous cultural heritage and preservation of cultural artifacts; prepared reports for the UNESCO committee.

Delegate of Israel, Model United Nations, Buenos Aires, Argentina — 09/2018 - 10/2018

- Conducted research on policies for reducing global climate change in developing countries.

Leadership Experience

Administrative Head, Luminous Life Foundation, Lagos, Nigeria — 09/2023 - Present

- Led a team of 15 to deploy power solutions to underserved homes without electricity for over 50 years in rural Nigeria.

Academic Officer, Dept. of Electrical and Electronics Engineering, Covenant University — 09/2022 - 08/2023

- Served as a liaison for 500+ students and 20 faculty members, ensuring healthy academic welfare.
- Coordinated tutorials and curated past questions in collaboration with TAG for students across all years.
- Organized symposiums with industry leaders to connect Electrical Engineering students to industries.

Volunteer Work & Community Service

Co-founder, Tech for Progress, Isolo, Lagos, Nigeria 01/2023 - Present

- Initiated and led Isolo's first digital training for underserved youth, equipping 50+ students with essential tech skills.

Career Advisor, Tesla's Academic Group (TAG), Covenant University 03/2022 - 03/2023

- Guided and supported 400+ students in their academic and career development through periodic webinars.

Lead AI Tutor, Google Developer Student Committee (GDSC), Covenant University 01/2022 - 12/2022

- Taught and expanded a community of AI enthusiasts from 200 to 500+ through custom curricula.
- Led the community to develop AI projects centered around NLP and Computer Vision for image detection.

Member Assistant Project Manager (National Champions), Enactus, Covenant University11/2019 - 09/2022

- Led the development of ECOBAG, a heat retention cooler for SMEs, to combat unstable power supply and reduce spoilage.
- Assisted in developing PET-City, a plastics-to-bricks scheme, to reduce plastic waste.
- Achieved National Champions status among 18 teams nationwide in 2020 for the impact of PET-City.

Tutor, Deeper Life Student Outreach (DLSO), Ojo, Lagos, Nigeria 07/2019 - 08/2019

- Prepared 20+ students for college entrance exams while guiding their career choices.

Skills

- **Programming Languages:** Python, C, C++, Javascript, SQL.
- **Frameworks & Libraries:** Flask, ROS, Gazebo, OpenCV.
- **AI & Deep Learning:** Data analysis with Matplotlib, NumPy, PyTorch, scikit-learn, TensorFlow, seaborn.
- **Embedded Systems:** Microcontrollers, sensors, I2C, SPI, UART, soldering, KiCad.
- **Tooling & CI/CD:** Git, GitHub.
- **Soft Skills:** Writing; teaching; project, people, and time management; leadership; critical thinking.

Selected Projects

CamAI: **Action Recognition Camera System**, MediaPipe, Conv-LSTM, Computer Vision 03/2023 - 05/2023

- Managed and led a team of two in designing an action recognition system for patient monitoring in healthcare.
- Used MediaPipe for feature extraction from the camera's video feed and an AI ensemble model (Conv-LSTM) to predict the individual's state, while displaying the feed on a webpage.

Quad-X: **Quadcopter Flight Controller**, ATMEGA328-PU, PID, Sensor Fusion 01/2021 - 12/2021

- Designed and constructed a quadcopter flight controller using an ATMEGA328-PU microcontroller, implementing a PID algorithm and sensor fusion on IMU data for all-axis stabilization.

MilliMeter: **Low-Cost IoT Metering System**, Embedded Communication Protocols, Web 03/2020 - 07/2020

- Led a team of 4 to design a low-cost metering system and IoT solution using custom embedded communication protocols and web technologies for an in-state IoT smart-metering competition.

Honors & Awards

- **Fitzcelerate**, £2500, University of Cambridge – Runner-Up (2026).
- **Mastercard Foundation Scholar**, University of Cambridge – 69 of 4,218 (2025).
- **Best Graduating Student**, Covenant University – 1 of 1,175 (2023).
- **Best Graduating Student in Engineering & EIE**, Covenant University – 1 of 132 (2023).
- **Highest CGPA (4.98/5.0)** in the EIE Department's history at Covenant University (2023).
- **Dean's Award for Academic Excellence**, Covenant University – 1 of 386 (2023).
- **Femi Ademiju Award for Excellence** to the Best Graduating EIE Student (2023).
- **CUALA Award for Best Graduating Student**, College of Engineering, Covenant University (2023).
- **First Runner-Up**, 3-line Finance Hackathon (2022–2023).
- **First Runner-Up**, Innovation Hackathon, Covenant University (2022).
- **$100 Prize**, OMGRobotics Challenge, Goodwall (2021).
- **$250 Prize**, My Passion Challenge, Goodwall (2021).
- **Qualifier**, Lagos State Smart Meter Hackathon (2020).
- **Winner**, Thinkerthon (2020).
- **First Runner-Up**, Pan-African Hackathon (2020).
- **Best Student in IGCSE** and in Chemistry & Geography Studies, DLHS, Lagos (2018).

Media Appearances

- Best Graduating Student in Covenant University – featured (09/2023).
- 3line's Hackathon – featured (04/2023).